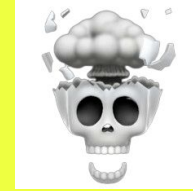


# **Палеогенетическое исследование дневного ячменя в домашних условиях**

## **или наш необычный опыт участия в агрохакатоне**

Стафеев Иван К3321 ТЭС 1.2

В ноябре 2024 мы поучаствовали в агрохакатоне от Россельхозбанка в одном из треков. Оказалось, что он по палеогенетике...



Нам дали такие вводные данные



>LT727735.1 Fragment Gene *Btr1*

```
atggcgagccgcccaatggaagcgcatgtaccagatgtgacgacggcgacgacgctgcgcccgcctgcagaaaagcgtgccgcgcgcgcgagcctgagcagcccgatgggtctggacacccgacgacccggcggtgcacgttgctcattccggtgacccacgtcagacgacatccgactgcctctccggttcatagtcagctgggtggtggcgagctcctgtgctccatggctgcggggccgtcccgtagggccggtggccagcatcgacggcctccggcaaccgcgacgaccacgacgagtgctcgtctgagcaggtcagggcccgaggagcagcgccaggagcgcgtcccgaggtggagggggccttcacccctcctggcctccgtccggttcagcttcgcagccggaccgccgacgcccggcgccgcaagccatggaagagcagctccacgcccgcgctcgaacttcaggccgtgggtggcgagcgtggcgaaacatgtccgctggcttcttggccactcagcctgccatccgaaccgcatccagtgag
```

>LT727751.1 Fragment Gene *Btr2*

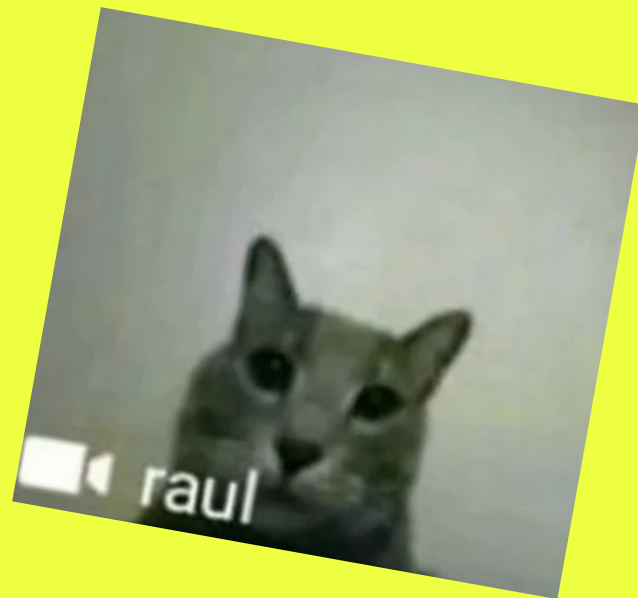
```
atggaggtttggaggaagacggcgccggagggcgtcggtcgtagcttgacctaatacaacggcagcaaatgcagtagtggaaggccatcatgccgcacgccagcggttacgcccctggcctccgaggactgccgccattccgccggcggtgcatccccgcccaacggcgccaggcgccctcagcaggcgcgacttcgtcagctcgctatcgaccggatcaagcgcttcagcaggttccagggcgtgctgggcaacgtctctccctctgcgctgcgctgcgcatcgcggtccaggtccaggccaaagcgccgctggcgccacaggtggcaactccaccagccgacgctgcccgccacgcgagacggcgctgactgcctgcactcggcgaaagtcgacggccatgcggccctggcgctctccagtcgatgctcagccgcatcgccgcgagcattggcccatgcctggcgcccgccgcggcgagcagctcctgcagcgcgtgatggacgactggccatggcggaaggccgcggtggagcgggatgcgccccgccattgtgccagttcttcgacgcctcgtgatgtcttcgatgctga
```

>AB489121.1 Fragments Gene *Vrs1*

```
atggacaagcatcagctctttgattcatccaagctggacacgactttcttcgcggccaatg.....gcacggcgagggggataccagcaagcagagggcgcgcgacggcgccgaggtgcgaggtgcggcgaggggatgggtgacgggtggggagatggacgagggaggggaccccaagaagcggcggtcaccgagcagcagcccgagattctggagctgagcttccgggagaccgcaagctggagacagcccgaaggtgtatcttgcccgcgagctcggctggaccccaagcaggtcccggtgtgtccagaaccgcccgcgcgccacaagaacaagacgctcgaggaggagttcgcgaggtcgaagcagccccagacgcccacatctccaaaatgccacctcgagaacgag...ctgctgaggctgaaggagagactgggagcgactgagcagagggtcgcgccctcaggctggcagctgggagccacgggcatctgtggatggcgacacgcccgtcgccgctgtgcgtgtgcggcgagcccgagctcgtcttcgacgggaacctgccagcagcagccgggtttcagcggggcagacgtgctcgggcgacgatgactgatgtgctgctcccgagtggttttagcatga
```

И дали вопрос, на который нужно ответить.

**“Выращивали ли из этого ячменя на Руси пиво??”**



мы с командой ЭТОТ МОМЕНТ

## Разбираемся с генами



У нас фрагменты трех генов: BTR1, BTR2, VRS1

**BTR1** и **BTR2** отвечают за ломкость колоса ячменя: если оба доминантные (**Btr1**, **Btr2**), то колос ломкий, если хотя бы один рецессивный (**btr1**, **btr2**), то колос стойкий

**VRS1** отвечает за дву- или шестирядность колоса. Если доминантный, то колос двурядный



## Выравнивание генов

Мы взяли несколько изученных ДНК других ячменей, как **домашних**, так и **диких**, и начали сравнивать их последовательности ДНК

У одомашненных неломкоколосных SBCC107 и SBCC043 наблюдается мутация: произошла делеция цитозина в экзона в позиции 272 пн. Здесь рецессивный аллельный ген **btr1**

У нашего ячменя делеции нет, значит, у нас **Btr1**

### LT727735.1 Fragment Gene Btr1

LT727735.1 *Hordeum vulgare* cv. SBCC145, allele Btr1 (reference)

LT727732.1 *Hordeum spontaneum* cv. HS6, allele Btr1

LT727730.1 *Hordeum spontaneum* cv. HS1, allele Btr1

LT727733.1 *Hordeum vulgare* cv. SBCC073, allele Btr1

LT727746.1 *Hordeum vulgare* cv. SBCC107, allele btr1

LT727745.1 *Hordeum vulgare* cv. SBCC043, allele btr1

|     | 254 | 256 | 258 | 260 | 262 | 264 | 266 | 268 | 270 | 272 | 274 | 276 | 278 | 280 | 282 | 284 | 286 | 288 | 291 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|-----|
| 182 | T   | C   | G   | A   | G   | C   | A   | C   | G   | C   | A   | T   | C   | C   | G   | A   | C   | T   | G   | C | C | T | C | T | C | C | G | G | T | T | C | A | T | A | G | T | C | A | C | 222 |
| 253 | T   | C   | G   | A   | G   | C   | A   | C   | G   | C   | A   | T   | C   | C   | G   | A   | C   | T   | G   | C | C | T | C | T | C | C | G | G | T | T | C | A | T | A | G | T | C | A | C | 293 |
| 247 | T   | C   | G   | A   | G   | C   | A   | C   | G   | C   | A   | T   | C   | C   | G   | A   | C   | T   | G   | C | C | T | C | T | C | C | G | G | T | T | C | A | T | A | G | T | C | A | C | 287 |
| 245 | T   | C   | G   | A   | G   | C   | A   | C   | G   | C   | A   | T   | C   | C   | G   | A   | C   | T   | G   | C | C | T | C | T | C | C | G | G | T | T | C | A | T | A | G | T | C | A | C | 285 |
| 203 | T   | C   | G   | A   | G   | C   | A   | C   | G   | C   | A   | T   | C   | C   | G   | A   | C   | T   | G   | C | C | T | C | T | C | C | G | G | T | T | C | A | T | A | G | T | C | A | C | 243 |
| 199 | T   | C   | G   | A   | G   | C   | A   | C   | G   | C   | A   | T   | C   | C   | G   | A   | C   | T   | G   | - | C | T | C | T | C | C | G | G | T | T | C | A | T | A | G | T | C | A | C | 238 |
| 199 | T   | C   | G   | A   | G   | C   | A   | C   | G   | C   | A   | T   | C   | C   | G   | A   | C   | T   | G   | - | C | T | C | T | C | C | G | G | T | T | C | A | T | A | G | T | C | A | C | 238 |

# Выравнивание генов

| LT727751.1 Fragment Gene Btr2 |                                                                 | 326 | 328 | 330 | 332 | 334 | 336 | 338 | 340 | 342 | 344 | 346 | 348 | 350 | 352 | 354 | 356 | 358 | 360 | 362 | 365 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |     |
|-------------------------------|-----------------------------------------------------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|-----|-----|
| 239                           |                                                                 | C   | C   | A   | G   | G   | C   | C   | G   | T   | G   | C   | T   | G   | G   | G   | C   | A   | A   | C   | G   | T | C | T | T | C | T | C | C | C | T | C | T | G | C | G | C | T | G | C | G | 281 |     |
| 270                           | LT727751.1 Hordeum vulgare cv. SBCC153, allele Btr2 (reference) | C   | C   | A   | G   | G   | C   | C   | G   | T   | G   | C   | T   | G   | G   | G   | C   | A   | A   | C   | G   | T | C | T | T | C | T | C | C | C | T | C | T | G | C | G | C | T | G | C | G | 312 |     |
| 270                           | LT727755.1 Hordeum spontaneum cv. HS6, allele Btr2              | C   | C   | A   | G   | G   | C   | C   | G   | T   | G   | C   | T   | G   | G   | G   | C   | A   | A   | C   | G   | T | C | T | T | C | T | C | C | C | T | C | T | G | C | G | C | T | G | C | G | 312 |     |
| 270                           | LT727753.1 Hordeum spontaneum cv. HS1, allele Btr2              | C   | C   | A   | G   | G   | C   | C   | G   | T   | G   | C   | T   | G   | G   | G   | C   | A   | A   | C   | G   | T | C | T | T | C | T | C | C | C | T | C | T | G | C | G | C | T | G | C | G | 312 |     |
| 270                           | LT727760.1 Hordeum vulgare cv. SBCC107, allele Btr2             | C   | C   | A   | G   | G   | C   | C   | G   | T   | G   | C   | T   | G   | G   | G   | C   | A   | A   | C   | G   | T | C | T | T | C | T | C | C | C | T | C | T | G | C | G | C | T | G | C | G | 312 |     |
| 270                           | LT727759.1 Hordeum vulgare cv. SBCC043, allele Btr2             | C   | C   | A   | G   | G   | C   | C   | G   | T   | G   | C   | T   | G   | G   | G   | C   | A   | A   | C   | G   | T | C | T | T | C | T | C | C | C | T | C | T | G | C | G | C | T | G | C | G | 312 |     |
| 325                           | LT727761.1 Hordeum vulgare cv. SBCC039, allele btr2             | C   | C   | A   | G   | G   | C   | C   | G   | T   | G   | C   | T   | G   | G   | -   | -   | -   | -   | -   | -   | - | - | - | - | - | - | T | C | C | C | T | C | T | G | C | G | C | T | G | C | G   | 356 |

У одомашненного неломкоколосного есть делеция в 11пн на позиции 340-350 пн. У ломкоколосных домашних и диких такой делеции нет, и они несут доминантный **Btr2**. Значит, у нашего тоже **Btr2**

**Промежуточно имеем, что наш ячмень был ломкоколосный**

## Выравнивание генов

С геном VRS все посложнее. В нашем распоряжении **10** аллелей: Vrs1.b1, Vrs1.b2, Vrs1.b3, Vrs1.b5, vrs1.a1, vrs1.a2, vrs1.a3, Vrs1.t1, Vrs1.t2 и Vrs1.t3

Но известно, что **vrs1.a2** и **vrs1.a3** произошли вследствие мутации от доминантных генов **Vrs1.b2** и **Vrs1.b3** соответственно, а Vrs.t еще не успел мутировать



# Выравнивание генов

AB489121.1 Fragments Gene Vrs1  
 AB489121.1 *Hordeum vulgare* cv. Bonus, allele Vrs1.b3 (reference)  
 AB489128.1 *Hordeum spontaneum* str. OUH602, allele Vrs1.b1  
 LT727701.1 *Hordeum spontaneum* cv. HS2, allele Vrs1.b2  
 AB259783.1 *Hordeum vulgare* Vrs1 for HD-ZIP, allele Vrs1.b1  
 LT727703.1 *Hordeum vulgare* cv. PI356226, allele Vrs1.b2  
 LT727707.1 *Hordeum vulgare* cv. SBCC154, allele Vrs1.b3  
 LT727716.1 *Hordeum vulgare* cv. SBCC153, allele Vrs1.b5  
 LT727705.1 *Hordeum vulgare* cv. Beatrix, allele Vrs1.t1  
 LC216330.1 *Hordeum vulgare* str. Deficiens 2, allele Vrs1.t2  
 LC216328.1 *Hordeum vulgare* cv. Semi-deficiens 12, allele Vrs1.t3  
**LT727697.1 *Hordeum vulgare* cv. SBCC115, allele vrs1.a1**  
 LT727718.1 *Hordeum vulgare* cv. SBCC039, allele vrs1.a2  
 EU331662.1 *Hordeum vulgare* cv. 6B94-8253, allele vrs1.a3

|      | 2414 | 2420 | 2425 | 2430 | 2435 | 2440 | 2445 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |
|------|------|------|------|------|------|------|------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|------|
| 439  | A    | G    | A    | C    | T    | G    | G    | G | A | G | C | G | A | C | T | G | A | G | C | A | G | G | A | G | G | T | G | C | G | G | C | G | C | 473  |
| 1335 | A    | G    | A    | C    | T    | G    | G    | G | A | G | C | G | A | C | T | G | A | G | C | A | G | G | A | G | G | T | G | C | G | G | C | G | C | 1369 |
| 1335 | A    | G    | A    | C    | T    | G    | G    | G | A | G | C | A | A | C | T | G | A | G | C | A | G | G | A | G | G | T | G | C | G | G | C | G | C | 1369 |
| 850  | A    | G    | A    | C    | T    | G    | G    | G | A | G | C | G | A | C | T | G | A | G | C | A | G | G | A | G | G | T | G | C | G | G | C | G | C | 884  |
| 591  | A    | G    | A    | C    | T    | G    | G    | G | A | G | C | G | A | C | T | G | A | G | C | A | G | G | A | G | G | T | G | C | G | G | C | G | C | 625  |
| 850  | A    | G    | A    | C    | T    | G    | G    | G | A | G | C | G | A | C | T | G | A | G | C | A | G | G | A | G | G | T | G | C | G | G | C | G | C | 884  |
| 850  | A    | G    | A    | C    | T    | G    | G    | G | A | G | C | G | A | C | T | G | A | G | C | A | G | G | A | G | G | T | G | C | G | G | C | G | C | 884  |
| 850  | A    | G    | A    | C    | T    | G    | G    | G | A | G | C | G | A | C | T | G | A | G | C | A | G | G | A | G | G | T | G | C | G | G | C | G | C | 884  |
| 850  | A    | G    | A    | C    | T    | G    | G    | G | A | G | C | G | A | C | T | G | A | G | C | A | G | G | A | G | G | T | G | C | G | G | C | G | C | 884  |
| 2412 | A    | G    | A    | C    | T    | G    | G    | G | A | G | C | G | A | C | T | G | A | G | C | A | G | G | A | G | G | T | G | C | G | G | C | G | C | 2446 |
| 2412 | A    | G    | A    | C    | T    | G    | G    | G | A | G | C | G | A | C | T | G | A | G | C | A | G | G | A | G | G | T | G | C | G | G | C | G | C | 2446 |
| 850  | A    | G    | A    | C    | T    | G    | G    | G | A | G | C | G | A | C | T | G | A | - | C | A | G | G | A | G | G | T | G | C | G | G | C | G | C | 883  |
| 851  | A    | G    | A    | C    | T    | G    | G    | G | A | G | C | G | A | C | T | G | A | G | C | A | G | G | A | G | G | T | G | C | G | G | C | G | C | 885  |
| 901  | A    | G    | A    | C    | T    | G    | G    | G | A | G | C | G | A | C | T | G | A | G | C | A | G | G | A | G | G | T | G | C | G | G | C | G | C | 935  |

У **vrs1.a1** происходит делеция гуанина в 2430пн, у нашего ячменя такого нет



# Выравнивание генов

AB489121.1 Fragments Gene Vrs1  
 AB489121.1 Hordeum vulgare cv. Bonus, allele Vrs1.b3 (reference)  
 AB489128.1 Hordeum spontaneum str. OUH602, allele Vrs1.b1  
 LT727701.1 Hordeum spontaneum cv. HS2, allele Vrs1.b2  
 AB259783.1 Hordeum vulgare Vrs1 for HD-ZIP, allele Vrs1.b1  
 LT727703.1 Hordeum vulgare cv. PI356226, allele Vrs1.b2  
 LT727707.1 Hordeum vulgare cv. SBCC154, allele Vrs1.b3  
 LT727716.1 Hordeum vulgare cv. SBCC153, allele Vrs1.b5  
 LT727705.1 Hordeum vulgare cv. Beatrix, allele Vrs1.t1  
 LC216330.1 Hordeum vulgare str. Deficiens 2, allele Vrs1.t2  
 LC216328.1 Hordeum vulgare cv. Semi-deficiens 12, allele Vrs1.t3  
 LT727697.1 Hordeum vulgare cv. SBCC115, allele vrs1.a1  
**LT727718.1 Hordeum vulgare cv. SBCC039, allele vrs1.a2**  
 EU331662.1 Hordeum vulgare cv. 6B94-8253, allele vrs1.a3

|      | 1 974 |   |   |   | 1980 |   |   |   | 1985 |   |   |   | 1990 |   |   |   | 1995 |   |   |   | 2k |   |   |   | 2 005 |   |   |   |   |   |   |   |      |
|------|-------|---|---|---|------|---|---|---|------|---|---|---|------|---|---|---|------|---|---|---|----|---|---|---|-------|---|---|---|---|---|---|---|------|
| 101  | G     | C | A | G | G    | C | G | G | C    | G | G | A | G    | G | T | C | G    | G | C | - | G  | A | G | G | T     | G | C | G | G | C | G | A | 134  |
| 896  | G     | C | A | G | G    | C | G | G | C    | G | G | A | G    | G | T | C | G    | G | C | - | G  | A | G | G | T     | G | C | G | G | C | G | A | 929  |
| 896  | G     | C | A | G | G    | C | G | G | C    | G | G | A | G    | G | T | C | G    | G | C | - | G  | A | G | G | T     | G | C | G | G | C | G | A | 929  |
| 411  | G     | C | A | G | G    | C | G | G | C    | G | G | A | G    | G | T | C | G    | G | C | - | G  | A | G | G | T     | G | C | G | G | C | G | A | 444  |
| 253  | G     | C | A | G | G    | C | G | G | C    | G | G | A | G    | G | T | C | G    | G | C | - | G  | A | G | G | T     | G | C | G | G | C | G | A | 286  |
| 411  | G     | C | A | G | G    | C | G | G | C    | G | G | A | G    | G | T | C | G    | G | C | - | G  | A | G | G | T     | G | C | G | G | C | G | A | 444  |
| 411  | G     | C | A | G | G    | C | G | G | C    | G | G | A | G    | G | T | C | G    | G | C | - | G  | A | G | G | T     | G | C | G | G | C | G | A | 444  |
| 410  | G     | C | A | G | G    | C | G | G | C    | G | G | A | G    | G | T | C | G    | G | C | T | G  | A | G | G | T     | G | C | G | G | C | G | A | 444  |
| 411  | G     | C | A | G | G    | C | G | G | C    | G | G | A | G    | G | T | C | G    | G | C | - | G  | A | G | G | T     | G | C | G | G | C | G | A | 444  |
| 1973 | G     | C | A | G | G    | C | G | G | C    | G | G | A | G    | G | T | C | G    | G | C | - | G  | A | G | G | T     | G | C | G | G | C | G | A | 2006 |
| 1973 | G     | C | A | G | G    | C | G | G | C    | G | G | A | G    | G | T | C | G    | G | C | - | G  | A | G | G | T     | G | C | G | G | C | G | A | 2006 |
| 411  | G     | C | A | G | G    | C | G | G | C    | G | G | A | G    | G | T | C | G    | G | C | - | G  | A | G | G | T     | G | C | G | G | C | G | A | 444  |
| 411  | G     | C | A | G | G    | C | G | G | C    | G | G | A | G    | G | T | C | G    | G | C | T | G  | A | G | G | T     | G | C | G | G | C | G | A | 445  |
| 462  | G     | C | A | G | G    | C | G | G | C    | G | G | A | G    | G | T | C | G    | G | C | - | G  | A | G | G | T     | G | C | G | G | C | G | A | 495  |

У **vrs1.a2** происходит вставка тимина на 1992пн, у нашего ячменя такого нет

# Выравнивание генов

AB489121.1 Fragments Gene Vrs1

AB489121.1 *Hordeum vulgare* cv. Bonus, allele Vrs1.b3 (reference)

AB489128.1 *Hordeum spontaneum* str. OUH602, allele Vrs1.b1

LT727701.1 *Hordeum spontaneum* cv. HS2, allele Vrs1.b2

AB259783.1 *Hordeum vulgare* Vrs1 for HD-ZIP, allele Vrs1.b1

LT727703.1 *Hordeum vulgare* cv. PI356226, allele Vrs1.b2

LT727707.1 *Hordeum vulgare* cv. SBCC154, allele Vrs1.b3

LT727716.1 *Hordeum vulgare* cv. SBCC153, allele Vrs1.b5

LT727705.1 *Hordeum vulgare* cv. Beatrix, allele Vrs1.t1

LC216330.1 *Hordeum vulgare* str. Deficiens 2, allele Vrs1.t2

LC216328.1 *Hordeum vulgare* cv. Semi-deficiens 12, allele Vrs1.t3

LT727697.1 *Hordeum vulgare* cv. SBCC115, allele vrs1.a1

LT727718.1 *Hordeum vulgare* cv. SBCC039, allele vrs1.a2

**EU331662.1 *Hordeum vulgare* cv. 6B94-8253, allele vrs1.a3**

|      | 2 082 | 2085 |   | 2090 |   | 2095 |   | 2.1k |   | 2105 |   | 2110 |   | 2 114 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |
|------|-------|------|---|------|---|------|---|------|---|------|---|------|---|-------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|------|
| 208  | T     | T    | C | T    | G | G    | A | G    | C | T    | G | A    | G | C     | T | T | C | C | G | G | G | A | G | G | A | C | C | G | C | A | A | G | C | 242  |
| 1003 | T     | T    | C | T    | G | G    | A | G    | C | T    | G | A    | G | C     | T | T | C | C | G | G | G | A | G | G | A | C | C | G | C | A | A | G | C | 1037 |
| 1003 | T     | T    | C | T    | G | G    | A | G    | C | T    | G | A    | G | C     | T | T | C | C | G | G | G | A | G | G | A | C | C | G | C | A | A | G | C | 1037 |
| 518  | T     | T    | C | T    | G | G    | A | G    | C | T    | G | A    | G | C     | T | T | C | C | G | G | G | A | G | G | A | C | C | G | C | A | A | G | C | 552  |
| 360  | T     | T    | C | T    | G | G    | A | G    | C | T    | G | A    | G | C     | T | T | C | C | G | G | G | A | G | G | A | C | C | G | C | A | A | G | C | 394  |
| 518  | T     | T    | C | T    | G | G    | A | G    | C | T    | G | A    | G | C     | T | T | C | C | G | G | G | A | G | G | A | C | C | G | C | A | A | G | C | 552  |
| 518  | T     | T    | C | T    | G | G    | A | G    | C | T    | G | A    | G | C     | T | T | C | C | G | G | G | A | G | G | A | C | C | G | C | A | A | G | C | 552  |
| 518  | T     | T    | C | T    | G | G    | A | G    | C | T    | G | A    | G | C     | T | T | C | C | G | G | G | A | G | G | A | C | C | G | C | A | A | G | C | 552  |
| 518  | T     | T    | C | T    | G | G    | A | G    | C | T    | G | A    | G | C     | T | T | C | C | G | G | G | A | G | G | A | C | C | G | C | A | A | G | C | 552  |
| 2080 | T     | T    | C | T    | G | G    | A | G    | C | T    | G | A    | G | C     | T | T | C | C | G | G | G | A | G | G | A | C | C | G | C | A | A | G | C | 2114 |
| 2080 | T     | T    | C | T    | G | G    | A | G    | C | T    | G | A    | G | C     | T | T | C | C | G | G | G | A | G | G | A | C | C | G | C | A | A | G | C | 2114 |
| 518  | T     | T    | C | T    | G | G    | A | G    | C | T    | G | A    | G | C     | T | T | C | C | G | G | G | A | G | G | A | C | C | G | C | A | A | G | C | 552  |
| 519  | T     | T    | C | T    | G | G    | A | G    | C | T    | G | A    | G | C     | T | T | C | C | G | G | G | A | G | G | A | C | C | G | C | A | A | G | C | 553  |
| 569  | T     | T    | C | T    | G | G    | A | G    | C | T    | G | A    | G | C     | T | T | G | C | G | G | G | A | G | G | A | C | C | G | C | A | A | G | C | 603  |

У **vrs1.a3** замена цитозина на гуанин на 2098пн, у нашего ячменя такого нет

**В целом, получается, VRS1 у него не рецессивный**



# Выравнивание генов

AB489121.1 Fragments Gene Vrs1  
 AB489121.1 *Hordeum vulgare* cv. Bonus, allele Vrs1.b3 (reference)  
 AB489128.1 *Hordeum spontaneum* str. OUH602, allele Vrs1.b1  
 LT727701.1 *Hordeum spontaneum* cv. HS2, allele Vrs1.b2  
 AB259783.1 *Hordeum vulgare* Vrs1 for HD-ZIP, allele Vrs1.b1  
 LT727703.1 *Hordeum vulgare* cv. PI356226, allele Vrs1.b2  
 LT727707.1 *Hordeum vulgare* cv. SBCC154, allele Vrs1.b3  
 LT727716.1 *Hordeum vulgare* cv. SBCC153, allele Vrs1.b5  
**LT727705.1 *Hordeum vulgare* cv. Beatrix, allele Vrs1.t1**  
**LC216330.1 *Hordeum vulgare* str. Deficiens 2, allele Vrs1.t2**  
**LC216328.1 *Hordeum vulgare* cv. Semi-deficiens 12, allele Vrs1.t3**  
 LT727697.1 *Hordeum vulgare* cv. SBCC115, allele vrs1.a1  
 LT727718.1 *Hordeum vulgare* cv. SBCC039, allele vrs1.a2  
 EU331662.1 *Hordeum vulgare* cv. 6B94-8253, allele vrs1.a3

|      | 2010 | 2020 | 2025 | 2030 | 2035 | 2040 | 2045 | 2050 | 2055 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |      |
|------|------|------|------|------|------|------|------|------|------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|-----|------|
| 538  | T    | G    | T    | G    | C    | G    | G    | A    | G    | C | C | G | A | G | C | T | C | T | C | T | C | G | A | 573 |      |
| 1434 | T    | G    | T    | G    | C    | G    | G    | A    | G    | C | C | G | A | G | C | T | C | T | C | T | T | C | G | A   | 1469 |
| 1434 | T    | G    | T    | G    | C    | G    | G    | A    | G    | C | C | G | A | G | C | T | C | T | C | T | T | C | G | A   | 1469 |
| 949  | T    | G    | T    | G    | C    | G    | G    | A    | G    | C | C | G | A | G | C | T | C | T | C | T | T | C | G | A   | 984  |
| 690  | T    | G    | T    | G    | C    | G    | G    | A    | G    | C | C | G | A | G | C | T | C | T | C | T | T | C | G | A   | 725  |
| 949  | T    | G    | T    | G    | C    | G    | G    | A    | G    | C | C | G | A | G | C | T | C | T | C | T | T | C | G | A   | 984  |
| 949  | T    | G    | T    | G    | C    | G    | G    | A    | G    | C | C | G | A | G | C | T | C | T | C | T | T | C | G | A   | 984  |
| 949  | T    | G    | T    | G    | C    | G    | G    | A    | G    | C | C | G | A | G | C | T | C | T | C | T | T | C | G | A   | 984  |
| 949  | T    | G    | T    | G    | C    | G    | G    | A    | G    | C | C | G | A | G | C | T | C | T | C | T | T | C | G | A   | 984  |
| 2511 | T    | G    | T    | G    | C    | G    | G    | A    | G    | C | C | G | A | G | C | T | C | T | C | T | T | C | G | A   | 2546 |
| 2511 | T    | G    | T    | G    | C    | G    | G    | A    | G    | C | C | G | A | G | C | T | C | T | C | T | T | C | G | A   | 2546 |
| 948  | T    | G    | T    | G    | C    | G    | G    | A    | G    | C | C | G | A | G | C | T | C | T | C | T | T | C | G | A   | 983  |
| 950  | T    | G    | T    | G    | C    | G    | G    | A    | G    | C | C | G | A | G | C | T | C | T | C | T | T | C | G | A   | 985  |
| 1000 | T    | G    | T    | G    | C    | G    | G    | A    | G    | C | C | G | A | G | C | T | C | T | C | T | T | C | G | A   | 1035 |

Сравнением  
нашли, что у  
нас **Vrs1.b3**

AB489121.1 Fragments Gene Vrs1  
 AB489121.1 *Hordeum vulgare* cv. Bonus, allele Vrs1.b3 (reference)  
**AB489128.1 *Hordeum spontaneum* str. OUH602, allele Vrs1.b1**  
**LT727701.1 *Hordeum spontaneum* cv. HS2, allele Vrs1.b2**  
**AB259783.1 *Hordeum vulgare* Vrs1 for HD-ZIP, allele Vrs1.b1**  
**LT727703.1 *Hordeum vulgare* cv. PI356226, allele Vrs1.b2**  
**LT727707.1 *Hordeum vulgare* cv. SBCC154, allele Vrs1.b3**  
**LT727716.1 *Hordeum vulgare* cv. SBCC153, allele Vrs1.b5**  
 LT727705.1 *Hordeum vulgare* cv. Beatrix, allele Vrs1.t1  
 LC216330.1 *Hordeum vulgare* str. Deficiens 2, allele Vrs1.t2  
 LC216328.1 *Hordeum vulgare* cv. Semi-deficiens 12, allele Vrs1.t3  
 LT727697.1 *Hordeum vulgare* cv. SBCC115, allele vrs1.a1  
 LT727718.1 *Hordeum vulgare* cv. SBCC039, allele vrs1.a2  
 EU331662.1 *Hordeum vulgare* cv. 6B94-8253, allele vrs1.a3

|      | 1 933 | 1940 |   |   |   |   | 1945 |   |   |   |   | 1950 |   |   |   |   | 1955 |   |   |   |   | 1960 |   |   |   |   | 1 964 |   |   |   |   |   |   |      |
|------|-------|------|---|---|---|---|------|---|---|---|---|------|---|---|---|---|------|---|---|---|---|------|---|---|---|---|-------|---|---|---|---|---|---|------|
| 61   | G     | G    | C | A | C | G | G    | C | G | C | A | G    | G | G | G | G | A    | T | A | C | C | A    | G | C | A | A | G     | C | A | G | A | G | G | 94   |
| 855  | G     | G    | C | A | C | G | G    | C | G | C | A | G    | G | G | G | G | A    | T | A | C | C | A    | G | C | A | A | G     | C | A | G | A | G | G | 889  |
| 855  | G     | G    | C | A | C | G | G    | C | G | C | A | G    | G | G | G | G | A    | G | A | C | C | A    | G | C | A | A | G     | C | A | G | A | G | G | 889  |
| 370  | G     | G    | C | A | C | G | G    | C | G | C | A | G    | G | G | G | G | A    | G | A | C | C | A    | G | C | A | A | G     | C | A | G | A | G | G | 404  |
| 213  | G     | G    | C | A | C | G | G    | C | G | C | A | G    | G | G | G | G | A    | T | A | C | C | A    | G | C | A | A | G     | C | A | G | A | G | G | 246  |
| 370  | G     | G    | C | A | C | G | G    | C | G | C | A | G    | G | G | G | G | A    | G | A | C | C | A    | G | C | A | A | G     | C | A | G | A | G | G | 404  |
| 370  | G     | G    | C | A | C | G | G    | C | G | C | A | G    | G | G | G | G | A    | T | A | C | C | A    | G | C | A | A | G     | C | A | G | A | G | G | 404  |
| 370  | G     | G    | C | A | C | G | G    | C | G | C | A | -    | G | G | G | G | A    | G | A | C | C | A    | G | C | A | A | G     | C | A | G | A | G | G | 403  |
| 370  | G     | G    | C | A | C | G | G    | C | G | C | A | G    | G | G | G | G | A    | G | A | C | C | A    | G | C | A | A | G     | C | A | G | A | G | G | 404  |
| 1932 | G     | G    | C | A | C | G | G    | C | G | C | A | G    | G | G | G | G | A    | T | A | C | C | A    | G | C | A | A | G     | C | A | G | A | G | G | 1966 |
| 1932 | G     | G    | C | A | C | G | G    | C | G | C | A | G    | G | G | G | G | A    | T | A | C | C | A    | G | C | A | A | G     | C | A | G | A | G | G | 1966 |
| 370  | G     | G    | C | A | C | G | G    | C | G | C | A | G    | G | G | G | G | A    | G | A | C | C | A    | G | C | A | A | G     | C | A | G | A | G | G | 404  |
| 370  | G     | G    | C | A | C | G | G    | C | G | C | A | G    | G | G | G | G | A    | G | A | C | C | A    | G | C | A | A | G     | C | A | G | A | G | G | 404  |
| 421  | G     | G    | C | A | C | G | G    | C | G | C | A | G    | G | G | G | G | A    | T | A | C | C | A    | G | C | A | A | G     | C | A | G | A | G | G | 455  |

## Итого

**Btr1** и **Btr2** доминантные. Колос ломкий  
**Vrs1** доминантный. Колос двурядный



Для пива не подходит,  
плохое сырье

А на Русь этот ячмень  
могли занести через  
торговый путь из Турции  
или Персии, где дикий  
ячмень возник



**Спасибо, что дотерпели до конца!**